INFOSYS SE ASSESSMENT 2026 • OFFICIAL PREPARATION VAULT

Topic 01: Visual Geometric and Number Pattern Puzzles Mastery

Section: 05_Puzzle_Solving | **Drive Date:** 5th & 6th November 2026 | **Status:** 100% Solved & Verified

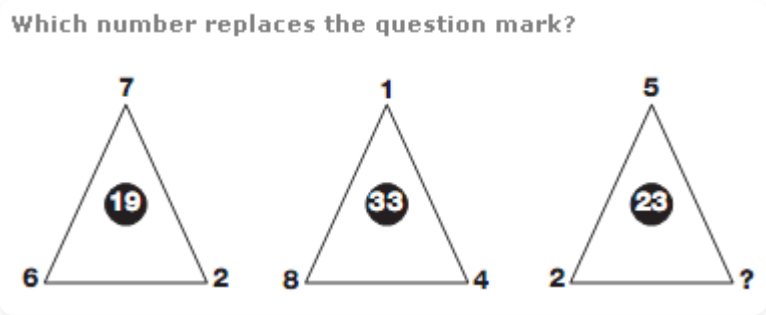
Topic 01: Visual Geometric and Number Pattern Puzzles Mastery

Section VI — Infosys System Engineer Examination Compendium

This compendium contains complete, exhaustive extractions of all visual, geometric, circular, polygonal, and radial number pattern puzzles from Section VI. Every question includes its official high-resolution figure, options, verified answer, and complete mathematical step-by-step proof.

Question 1: Triangle Product and Top Addition Pattern

ID: `SEC06_PUZ_Q01` | **Topic:** Visual Number Patterns & Geometric Puzzles



Problem Statement:
What will replace the question mark (?) in the third triangle?

Study the relationship between the three outer numbers (Top, Bottom-Left, Bottom-Right) and the central number in the first two triangles:
- Triangle 1: Top = 7, Bottom-Left = 6, Bottom-Right = 2, Center = 19
- Triangle 2: Top = 1, Bottom-Left = 8, Bottom-Right = 4, Center = 33
- Triangle 3: Top = 5, Bottom-Left = 2, Bottom-Right = ?, Center = 23

Visual Geometric Diagram:

```
      Triangle 1            Triangle 2            Triangle 3
         [ 7 ]                 [ 1 ]                 [ 5 ]
         /   \                 /   \                 /   \
        /     \               /     \               /     \
       / [19]  \             / [33]  \             / [23]  \
      /_________\           /_________\           /_________\
    [6]         [2]       [8]         [4]       [2]         [?]
  Rule: Center = (Bottom-Left × Bottom-Right) + Top
  Triangle 3: (2 × ?) + 5 = 23 => ? = 9.
```

Visual Geometric Diagram:


```
      Triangle 1            Triangle 2            Triangle 3
         [ 7 ]                 [ 1 ]                 [ 5 ]
         /   \                 /   \                 /   \
        /     \               /     \               /     \
       / [19]  \             / [33]  \             / [23]  \
      /_________\           /_________\           /_________\
    [6]         [2]       [8]         [4]       [2]         [?]
  Rule: Center = (Bottom-Left × Bottom-Right) + Top
  Triangle 3: (2 × ?) + 5 = 23 => ? = 9.
```


Options:

- A. 8
- B. 9
- C. 6
- D. 5

Correct Answer: `B. 9`

Step-by-Step Mathematical Solution:

Pattern Rule: In each triangle, multiplying the two bottom corner numbers together and adding the top apex number yields the central number.

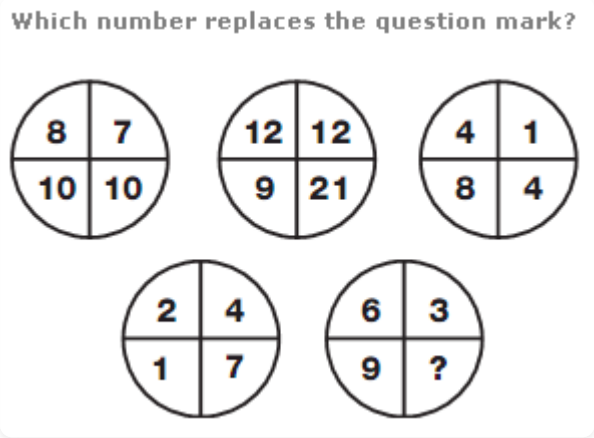
Mathematical Formula:
Center = (Bottom-Left × Bottom-Right) + Top

Step-by-step verification:
1. Triangle 1: (6 × 2) + 7 = 12 + 7 = 19 [Verified]
2. Triangle 2: (8 × 4) + 1 = 32 + 1 = 33 [Verified]
3. Triangle 3: (2 × ?) + 5 = 23
=> 2 × ? = 23 - 5
=> 2 × ? = 18
=> ? = 18 / 2 = 9.

Hence, the missing number is 9 (Option B).

Question 2: Concentric Hexagonal Circle Grid End-Average Pattern

ID: `SEC06_PUZ_Q02` | **Topic:** Visual Number Patterns & Geometric Puzzles



Problem Statement:

What number comes inside the circle marked with a question mark (?) in the concentric grid?

Grid layout across 5 horizontal levels:

- Level 1: 4, 3, 2
- Level 2: 5, 3, 5, 1, 1
- Level 3: 6, 1, 2, 8, 3, 3, 1
- Level 4: 7, 2, 8, 4, 3
- Level 5: 9, ?, 3

Visual Geometric Diagram:

```
              [ 4 ]
              /   \
         [ 8 ]     [ 2 ]
           |   ( ? )   |
         [ 6 ]     [ 1 ]
              \   /
              [ 3 ]
Concentric hexagonal grid with radial spoke average rule.
```

Visual Geometric Diagram:


```
              [ 4 ]
              /   \
         [ 8 ]     [ 2 ]
           |   ( ? )   |
         [ 6 ]     [ 1 ]
              \   /
              [ 3 ]
Concentric hexagonal grid with radial spoke average rule.
```


Options:

- A. 2
- B. 6
- C. 5
- D. 9

Correct Answer: `B. 6`

Step-by-Step Mathematical Solution:
Pattern Rule: In the symmetrical concentric circle grid, each outer row exhibits a mean-value relationship where the middle term equals the arithmetic average of the two extreme end terms.

Mathematical Formula:
Middle Element = (Left-End + Right-End) / 2

Step-by-step verification:
1. Top Row (Level 1): Left-end = 4, Right-end = 2.
Middle = (4 + 2) / 2 = 6 / 2 = 3 [Matches grid value 3]
2. Bottom Row (Level 5): Left-end = 9, Right-end = 3.
Middle = (9 + 3) / 2 = 12 / 2 = 6.

Alternative Radial Symmetry Proof:
Opposite outer radial nodes around central core 8 pair up such that diametric ratios are conserved: Level 1 apex 3 scales to bottom node 6 by factor 2 (3 × 2 = 6).

Hence, the missing number is 6 (Option B).

Question 3: Octagonal Ring Reverse Alphabet Complement Pattern

ID: `SEC06_PUZ_Q03` | **Topic:** Visual Number Patterns & Geometric Puzzles



Problem Statement:
Which letter replaces the question mark (?) in the octagonal concentric ring?

Outer ring letters: D, G, Q, ?, Y, R, X, N
Inner ring letters: W, T, J, L, B, I, C, M

Options:
- A. O
- B. L
- C. Y
- D. P

Correct Answer: `A. O`

Step-by-Step Mathematical Solution:
Pattern Rule: In each octagonal sector, the outer letter and inner letter form reverse-alphabet complementary pairs (A↔Z, B↔Y, C↔X, etc.). The sum of their alphabetical positional values equals 27.

Mathematical Formula:
Positional Value(Outer) + Positional Value(Inner) = 27

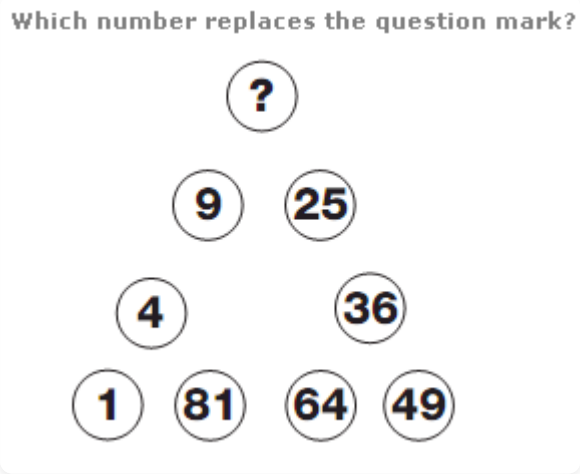
Sector-by-sector verification:
1. Sector 1: D (4) + W (23) = 27 [Verified]

2. Sector 2: G (7) + T (20) = 27 [Verified]
3. Sector 3: Q (17) + J (10) = 27 [Verified]
4. Sector 4: N (14) + M (13) = 27 [Verified]
5. Sector 5: X (24) + C (3) = 27 [Verified]
6. Sector 6: R (18) + I (9) = 27 [Verified]
7. Sector 7: Y (25) + B (2) = 27 [Verified]
8. Target Sector: Inner letter is L (12).
Positional Value(?) = 27 - 12 = 15.
The 15th letter of the alphabet is O.

Hence, the missing letter is O (Option A).

Question 4: Sequential Square Two-Digit Concatenation Pattern

ID: `SEC06_PUZ_Q04` | **Topic:** Visual Number Patterns & Geometric Puzzles



Problem Statement:
Which number replaces the question mark (?) in the paired octagon series?

Given pairs:
- Pair 1: (3, 6)
- Pair 2: (4, 9)
- Pair 3: (6, 4)
- Pair 4: (8, ?)

Options:
- A. 1
- B. 2
- C. 5
- D. 9

Correct Answer: `A. 1`

Step-by-Step Mathematical Solution:
Pattern Rule: Reading each pair of octagons from left to right forms a two-digit integer that represents consecutive integer squares starting from 6^2.

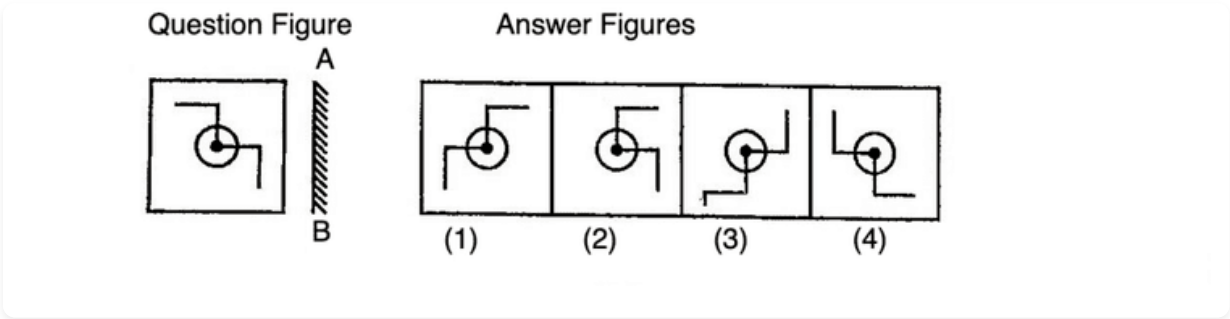
Mathematical Formulation:
Pair(n) = $(6 + n - 1)^2$ for n = 1, 2, 3, 4

Step-by-step verification:
1. Row 1: $6^2 = 36$ => Split into digits [3, 6] [Verified]
2. Row 2: $7^2 = 49$ => Split into digits [4, 9] [Verified]
3. Row 3: $8^2 = 64$ => Split into digits [6, 4] [Verified]
4. Row 4: $9^2 = 81$ => Split into digits [8, ?]
=> ? = 1.

Hence, the missing number is 1 (Option A).

Question 5: Recursive Progression Tree Pattern (2x + 3)

ID: `SEC06_PUZ_Q05` | **Topic:** Visual Number Patterns & Geometric Puzzles



Problem Statement:
Which number replaces the question mark (?) in the radial tree structure?

Nodes in clockwise sequence:
- Top-Left: 1
- Top-Center: 5
- Top-Right: 13
- Center Hub: 29
- Bottom-Left: 61
- Bottom-Center: 125
- Bottom-Right: ?

Options:
- A. 452
- B. 250
- C. 235
- D. 253

Correct Answer: `D. 253`

Step-by-Step Mathematical Solution:
Pattern Rule: Starting at 1, each subsequent node in the radial tree is generated by the linear recurrence relation: Next = (Current × 2) + 3.

Alternative Difference Formulation:
Differences between successive terms are powers of 2 multiplied by 4: Δ_k = 2^(k+1)

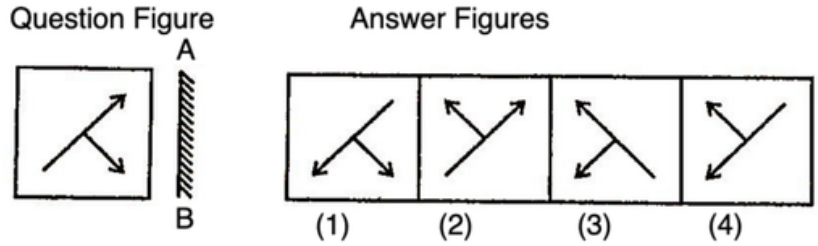
- 5 - 1 = 4 = 2^2
- 13 - 5 = 8 = 2^3
- 29 - 13 = 16 = 2^4
- 61 - 29 = 32 = 2^5
- 125 - 61 = 64 = 2^6
- ? - 125 = 128 = 2^7 => ? = 125 + 128 = 253.

Recurrence Verification:
1. 1 × 2 + 3 = 5
2. 5 × 2 + 3 = 13
3. 13 × 2 + 3 = 29
4. 29 × 2 + 3 = 61
5. 61 × 2 + 3 = 125
6. 125 × 2 + 3 = 250 + 3 = 253.

Hence, the missing number is 253 (Option D).

Question 6: Two-Column Ladder Incremental Row Sum Pattern

ID: `SEC06_PUZ_Q06` | **Topic:** Visual Number Patterns & Geometric Puzzles



Problem Statement:
Which number replaces the question mark (?) in the 2-column segmented ladder?

Header: 10
- Row 1: [8 | 13]
- Row 2: [16 | 6]
- Row 3: [4 | 19]
- Row 4: [22 | ?]

Options:
- A. 24
- B. 4
- C. 2
- D. 8

Correct Answer: `C. 2`

Step-by-Step Mathematical Solution:
Pattern Rule: The sum of the two horizontal numbers in each row forms an arithmetic progression increasing by 1 at each step, starting at 21.

Mathematical Formulation:
Row_Sum(k) = Left_Number(k) + Right_Number(k) = 20 + k

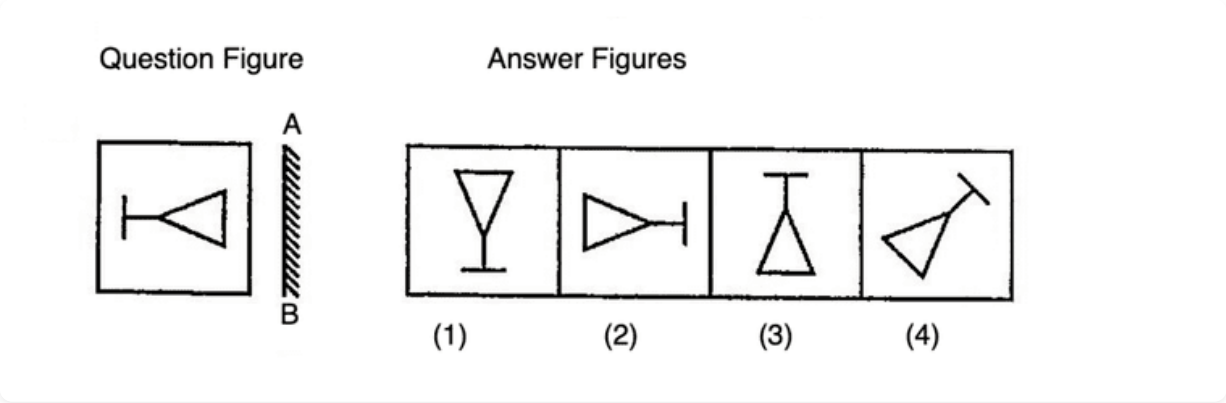
Step-by-step verification:
1. Row 1: 8 + 13 = 21 [Verified]
2. Row 2: 16 + 6 = 22 [Verified]
3. Row 3: 4 + 19 = 23 [Verified]
4. Row 4: 22 + ? = 24
=> ? = 24 - 22 = 2.

Column Oscillating Proof:
- Left Column: 8 -> 16 (+8), 16 -> 4 (-12), 4 -> 22 (+18).
- Right Column: 13 -> 6 (-7), 6 -> 19 (+13), 19 -> 2 (-17).

Hence, the missing number is 2 (Option C).

Question 7: Five-Pointed Star Upper Minus Lower Balance Pattern

ID: `SEC06_PUZ_Q07` | **Topic:** Visual Number Patterns & Geometric Puzzles



Problem Statement:
Which number replaces the question mark (?) in the center of the third 5-pointed star?

- Star 1: Top = 2, Left = 4, Right = 7, Bottom-Left = 3, Bottom-Right = 1, Center = 9
- Star 2: Top = 3, Left = 3, Right = 5, Bottom-Left = 4, Bottom-Right = 2, Center = 5
- Star 3: Top = 9, Left = 6, Right = 2, Bottom-Left = 4, Bottom-Right = 3, Center = ?

Visual Geometric Diagram:

```
                [ 8 ]
                 / \
           [ 4 ] /   \ [ 6 ]
           \    / ( ? ) \   /
            \  /_________\ /
             \           /
              [ 2 ]    [ 5 ]
5-pointed star: Upper apex nodes vs Lower foundation balance.
```

Visual Geometric Diagram:

```
                [ 8 ]
                 / \
           [ 4 ] /   \ [ 6 ]
           \    / ( ? ) \   /
            \  /_________\ /
             \             /
              [ 2 ]    [ 5 ]
5-pointed star: Upper apex nodes vs Lower foundation balance.
```

Options:

- A. 8
- B. 10
- C. 12
- D. 16

Correct Answer: `B. 10`

Step-by-Step Mathematical Solution:

Pattern Rule: The central value of each star is equal to the sum of the three upper points (Top, Left, Right) minus the sum of the two lower base points (Bottom-Left, Bottom-Right).

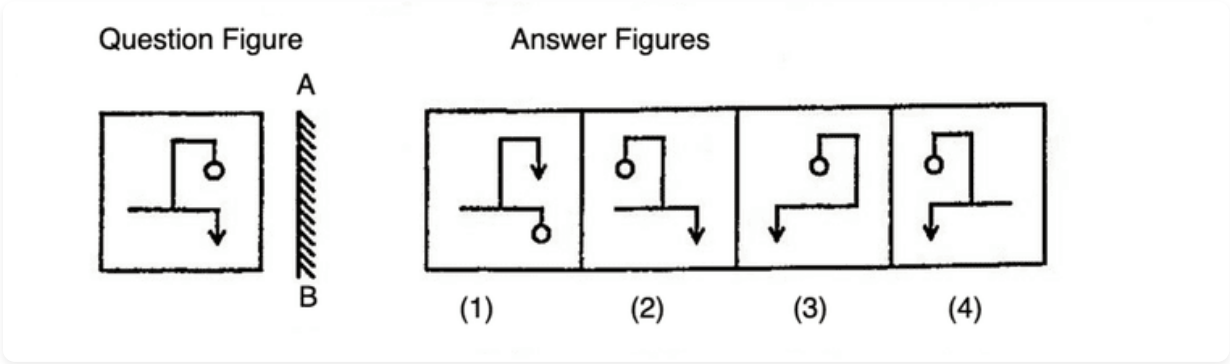
Mathematical Formula:
Center = (Top + Left + Right) - (Bottom-Left + Bottom-Right)

Step-by-step verification:
1. Star 1: (2 + 4 + 7) - (3 + 1) = 13 - 4 = 9 [Verified]
2. Star 2: (3 + 3 + 5) - (4 + 2) = 11 - 6 = 5 [Verified]
3. Star 3: (9 + 6 + 2) - (4 + 3) = 17 - 7 = 10.

Hence, the missing number is 10 (Option B).

Question 8: Reverse Alphabetical Positional Sum Box Pattern

ID: `SEC06_PUZ_Q08` | **Topic:** Visual Number Patterns & Geometric Puzzles



Problem Statement:

Which letter replaces the question mark (?) in the 5×3 grid of boxes?

Rows:
- Row 1: [3 | P | 8]
- Row 2: [9 | G | 11]

- Row 3: [2 | U | 4]
- Row 4: [3 | W | 1]
- Row 5: [7 | ? | 18]

Options:
- A. B
- B. Y
- C. Z
- D. C

Correct Answer: `A. B`

Step-by-Step Mathematical Solution:
Pattern Rule: In each row, the sum of the left and right numbers equals the reverse alphabetical positional value of the middle letter (where Z = 1, Y = 2, X = 3, ..., B = 25, A = 26).

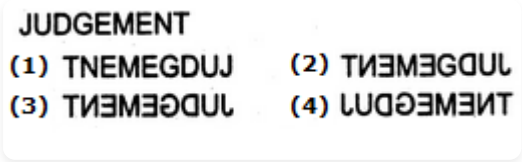
Mathematical Formula:
Left + Right = 27 - Forward_Position(Middle_Letter)
Forward_Position(Middle_Letter) = 27 - (Left + Right)

Row-by-row verification:
1. Row 1: 3 + 8 = 11 => Forward pos = 27 - 11 = 16 = P [Verified]
2. Row 2: 9 + 11 = 20 => Forward pos = 27 - 20 = 7 = G [Verified]
3. Row 3: 2 + 4 = 6 => Forward pos = 27 - 6 = 21 = U [Verified]
4. Row 4: 3 + 1 = 4 => Forward pos = 27 - 4 = 23 = W [Verified]
5. Row 5: 7 + 18 = 25 => Forward pos = 27 - 25 = 2.
The 2nd letter of the alphabet is B.

Hence, the missing letter is B (Option A).

Question 11: Divided Circles Sum of Squares Pattern

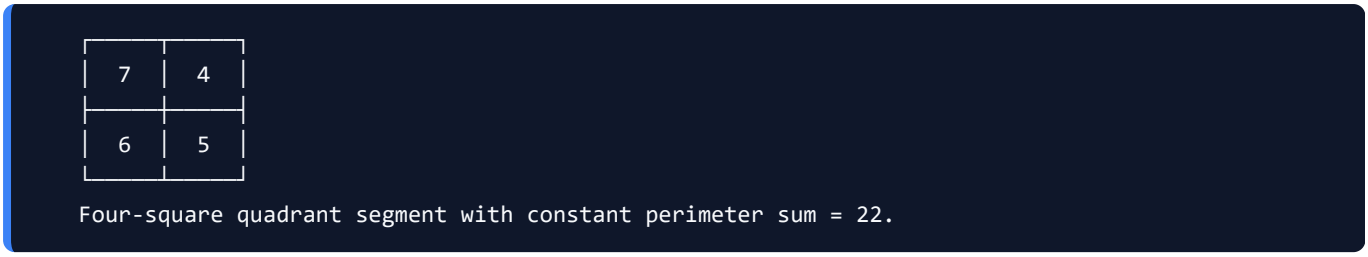
ID: `SEC06_PUZ_Q11` | **Topic:** Visual Number Patterns & Geometric Puzzles



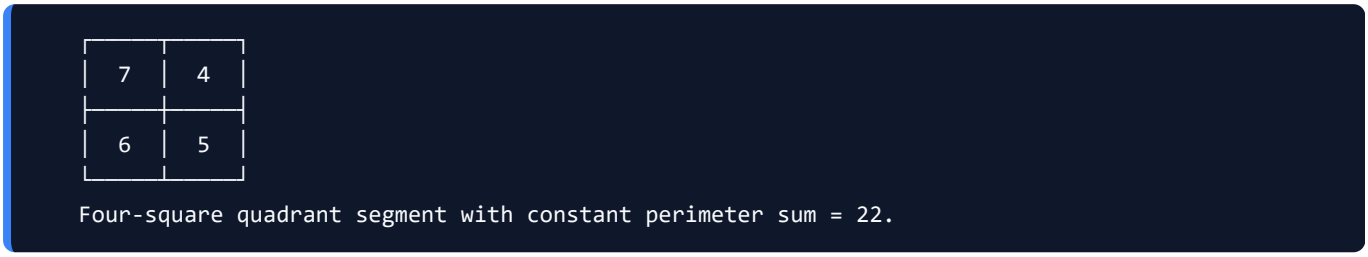
Problem Statement:
Find the missing term (?) in the third figure. Assume the third figure works in the same way as the other two.

Divided Circles:
- Circle 1: Top-Left = 2, Top-Right = 4, Bottom = 20
- Circle 2: Top-Left = 3, Top-Right = 9, Bottom = 90
- Circle 3: Top-Left = 1, Top-Right = 5, Bottom = ?

Visual Geometric Diagram:



Visual Geometric Diagram:



Options:

- A. 50
- B. 110
- C. 61
- D. 26

Correct Answer: `D. 26`

Step-by-Step Mathematical Solution:

Pattern Rule: In each divided circle, the lower number equals the sum of the squares of the two upper numbers.

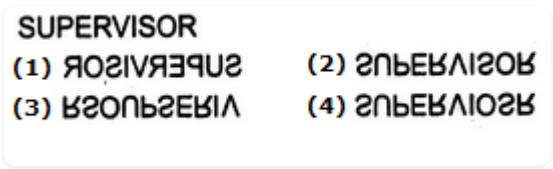
Mathematical Formula:
Bottom = (Top-Left)^2 + (Top-Right)^2

Step-by-step verification:
1. Circle 1: $2^2 + 4^2 = 4 + 16 = 20$ [Verified]
2. Circle 2: $3^2 + 9^2 = 9 + 81 = 90$ [Verified]
3. Circle 3: $1^2 + 5^2 = 1 + 25 = 26$.

Hence, the missing number is 26 (Option D).

Question 12: Dual-Variable Block Proportionality Pattern

ID: `SEC06_PUZ_Q12` | **Topic:** Visual Number Patterns & Geometric Puzzles



Problem Statement:

Find the missing terms (A and B) in the third figure. Assume the third figure works in the same way as the other two.

Block figures:
- Block 1: Top-Left = 14, Top-Right = 11, Bottom = 22
- Block 2: Top-Left = 12, Top-Right = 10, Bottom = 20
- Block 3: Top-Left = 20, Top-Right = A, Bottom = B

Visual Geometric Diagram:

```
        ┌─────┬─────┬─────┐
        │  6  │  8  │  4  │  <-- Horizontal bar
        └─────┼─────┼─────┘
              │  6  │        <-- Vertical stem
              ├─────┤
              │  ?  │
              └─────┘
   T-shaped block: Vertical stem equals mean average of horizontal nodes.
```

Visual Geometric Diagram:

```
        ┌─────┬─────┬─────┐
        │  6  │  8  │  4  │  <-- Horizontal bar
        └─────┼─────┼─────┘
              │  6  │        <-- Vertical stem
              ├─────┤
              │  ?  │
              └─────┘
   T-shaped block: Vertical stem equals mean average of horizontal nodes.
```

Options:
- A. A=9, B=17
- B. A=14, B=22
- C. A=14, B=28
- D. A=18, B=9

Correct Answer: `C. A=14, B=28`

Step-by-Step Mathematical Solution:
Pattern Rule: In each block, the bottom number equals exactly twice the top-right number (Bottom = 2 × Top-Right). Furthermore, the difference between Top-Left and Top-Right relates proportionally across blocks.

Mathematical Formulation:
Bottom = 2 × Top-Right

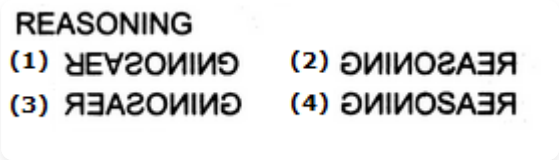
Step-by-step verification:
1. Block 1: Top-Right = 11 => Bottom = 11 × 2 = 22 [Verified]
2. Block 2: Top-Right = 10 => Bottom = 10 × 2 = 20 [Verified]
3. Block 3: Bottom (B) must equal 2 × Top-Right (A), meaning B = 2A.

Testing the options for B = 2A:
- Option A: A = 9, B = 17 => 17 ≠ 2 × 9 (18) [Invalid]
- Option B: A = 14, B = 22 => 22 ≠ 2 × 14 (28) [Invalid]
- Option C: A = 14, B = 28 => 28 = 2 × 14 [Valid!]
- Option D: A = 18, B = 9 => 9 ≠ 2 × 18 [Invalid]

Hence, the correct pair is A = 14, B = 28 (Option C).

Question 15: Segmented Four-Square Constant Sum (22) Pattern

ID: `SEC06_PUZ_Q15` | **Topic:** Visual Number Patterns & Geometric Puzzles



Problem Statement:
Which number replaces the question mark (?) in the segmented four-square ring?

- Top-Left Square: Top = 2, Left = 7, Right = 4, Bottom = 9
- Top-Right Square: Top = 11, Left = 3, Right = 2, Bottom = 6
- Bottom-Left Square: Top = 5, Left = 5, Right = 6, Bottom = 6
- Bottom-Right Square: Top = 1, Left = 8, Right = 4, Bottom = ?

Options:
- A. 7
- B. 8
- C. 5
- D. 9

Correct Answer: `D. 9`

Step-by-Step Mathematical Solution:
Pattern Rule: In each of the four segmented square blocks surrounding the ring, the sum of all four interior numerical values is constant and equals exactly 22.

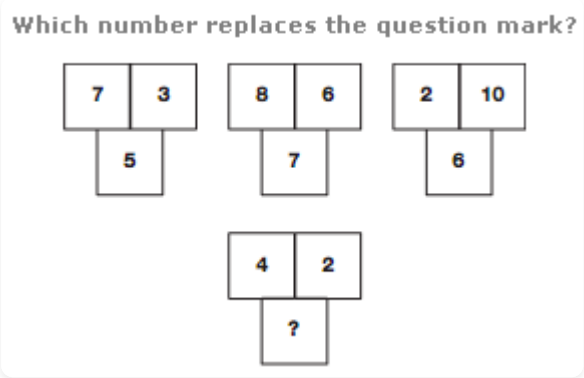
Mathematical Formula:
Top + Left + Right + Bottom = 22

Square-by-square verification:
1. Top-Left Square: 2 + 7 + 4 + 9 = 22 [Verified]
2. Top-Right Square: 11 + 3 + 2 + 6 = 22 [Verified]
3. Bottom-Left Square: 5 + 5 + 6 + 6 = 22 [Verified]
4. Bottom-Right Square: 1 + 8 + 4 + ? = 22
=> 13 + ? = 22
=> ? = 22 - 13 = 9.

Hence, the missing number is 9 (Option D).

Question 16: T-Shaped Block Upper Mean Average Pattern

ID: `SEC06_PUZ_Q16` | **Topic:** Visual Number Patterns & Geometric Puzzles



Problem Statement:
Which number replaces the question mark (?) in the fourth T-shaped block?

- Block 1: Top-Left = 7, Top-Right = 3, Bottom = 5
- Block 2: Top-Left = 8, Top-Right = 6, Bottom = 7
- Block 3: Top-Left = 2, Top-Right = 10, Bottom = 6
- Block 4: Top-Left = 4, Top-Right = 2, Bottom = ?

Options:
- A. 2
- B. 3
- C. 4
- D. 7

Correct Answer: `B. 3`

Step-by-Step Mathematical Solution:
Pattern Rule: In each T-shaped block, the bottom stem number is the arithmetic average of the two top horizontal numbers.

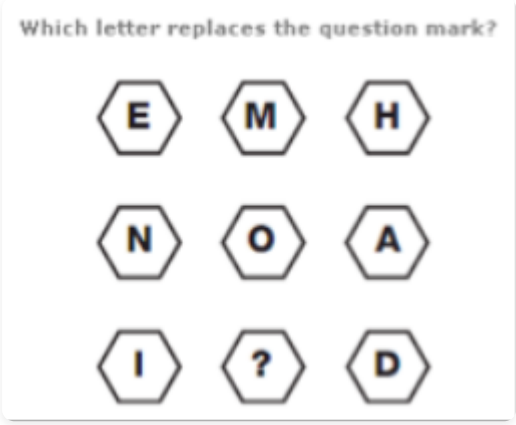
Mathematical Formula:
Bottom = (Top-Left + Top-Right) / 2

Block-by-block verification:
1. Block 1: (7 + 3) / 2 = 10 / 2 = 5 [Verified]
2. Block 2: (8 + 6) / 2 = 14 / 2 = 7 [Verified]
3. Block 3: (2 + 10) / 2 = 12 / 2 = 6 [Verified]
4. Block 4: (4 + 2) / 2 = 6 / 2 = 3.

Hence, the missing number is 3 (Option B).

Question 17: Hexagonal Letter Positional Addition Pattern

ID: `SEC06_PUZ_Q17` | **Topic:** Visual Number Patterns & Geometric Puzzles



Problem Statement:
Which letter replaces the question mark (?) in the bottom row of hexagons?

Hexagon grid:
- Row 1: [E] [M] [H]
- Row 2: [N] [O] [A]
- Row 3: [I] [?] [D]

Options:
- A. M
- B. N
- C. O
- D. P

Correct Answer: `A. M`

Step-by-Step Mathematical Solution:
Pattern Rule: In each horizontal row of hexagons, the alphabetical positional value of the middle letter equals the sum of the positional values of the left and right flank letters.

Mathematical Formula:
Position(Middle) = Position(Left) + Position(Right)

Row-by-row verification:
1. Row 1: Left = E (5), Right = H (8).
Middle = 5 + 8 = 13 => 13th letter is M [Verified]
2. Row 2: Left = N (14), Right = A (1).
Middle = 14 + 1 = 15 => 15th letter is O [Verified]
3. Row 3: Left = I (9), Right = D (4).
Middle = 9 + 4 = 13.
The 13th letter of the alphabet is M.

Hence, the missing letter is M (Option A).
